

Problem I : Probability

1)

$$P(\text{first is blue}) = \frac{b}{b+y}$$

$P(\text{second is blue}) = P(\text{2nd blue} \mid \text{1st blue}) * P(\text{1st blue}) + P(\text{2nd blue} \mid \text{1st yellow}) * P(\text{1st yellow})$

$$\begin{aligned} &= \frac{b-1}{b+y-1} * \frac{b}{b+y} + \frac{b}{b+y-1} * \frac{y}{b+y} \\ &= \frac{(b-1)b + by}{(b+y-1)(b+y)} = \frac{b^2 - b + by}{(b+y-1)(b+y)} = \frac{b(b-1+y)}{(b+y-1)(b+y)} = \frac{b}{(b+y)} \end{aligned}$$

2)

Let C = "A's answer was correct" and Y = "B says yes." We want $P(C \mid Y)$

Device A

Correct $P(C) : \frac{1}{3}$

Incorrect $P(C') : \frac{2}{3}$

Device B

Correct $P(Y \mid C) : \frac{1}{3}$

Incorrect $P(Y \mid C') : \frac{2}{3}$

Bayes Theorem:

$$P(C \mid Y) = \frac{P(Y \mid C) P(C)}{P(Y \mid C) P(C) + P(Y \mid C') P(C')} = \frac{\frac{1}{3} * \frac{1}{3}}{\frac{1}{3} * \frac{1}{3} + \frac{2}{3} * \frac{2}{3}} = \frac{1}{5}$$

3)

Two possibilities for envelope's original content:

H_p: original card is Premium is $\frac{1}{2}$

H_s: original card is Standard is $\frac{1}{2}$

With H_p: $P(\text{draw P} \mid H_p) = 1$

With H_s: $P(\text{draw P} \mid H_s) = \frac{1}{2}$

Bayes Theorem:

$$P(H_p \mid \text{draw P}) = \frac{1 * \frac{1}{2}}{1 * \frac{1}{2} + \frac{1}{2} * \frac{1}{2}} = \frac{2}{3}$$

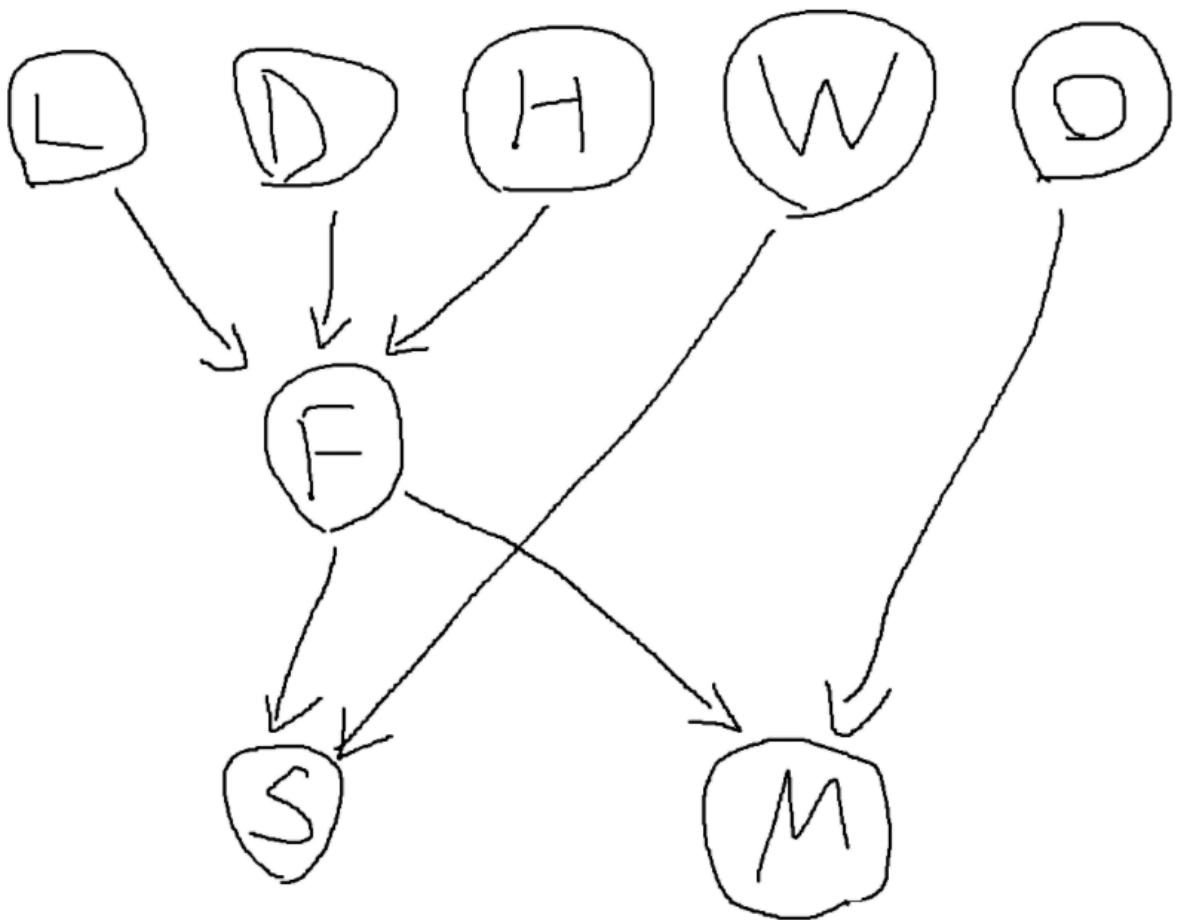
Given H_p , the remaining card is P with certainty and given H_s , the remaining card is S with certainty. Thus:

$$P(\text{remaining is P} \mid \text{drew P}) = \frac{2}{3} * 1 + \frac{1}{3} * 0 = \frac{2}{3}$$

Problem II : The Bayesian Network

Part 1:

1)



2) $P(L, D, H, W, O, F, S, M) =$

$$P(L) * P(D) * P(H) * P(W) * P(O) * P(F \mid L, D, H) * P(S \mid W, F) * P(M \mid F, O)$$

3)

Based on my Drawing, the ones that are independent from H are L, D, W, and O.

4)

(L,D), (L,H), (L,W), (L,O), (D,H), (D,W), (D,O), (H,W), (H,O), (W,O)

These are independent because there is no path at all between any two root nodes (no parents).

Part 1:

1)

Reading from the Wildfire table

Wildfire: $P(T) = 0.85$ $P(F) = 0.15$

2)

$P(\text{Lightning}=T \mid \text{WildFire}=T, \text{DryConditions}=T)$

Problem II Part 2 Joint Probability Computation:

1. $P(\text{WildFire} \mid H=T, L=T, D=T)$

Parents of WildFire match the evidence exactly.

Read the CPT row (T,T,T): $P(\text{WildFire}=T \mid \dots) = 0.85$, $P(\text{WildFire}=F \mid \dots) = 0.15$.

2. $P(\text{Lightning}=T \mid \text{WildFire}=T, \text{DryConditions}=T)$

$P(L=T \mid F=T, D=T) = P(L=T, F=T, D=T) / P(F=T, D=T)$.

Numerator: $P(L=T)P(D=T) \sum_h P(H=h) P(F=T \mid L=T, D=T, H=h) = 0.08 * 0.45 *$

$(0.09*0.85 + 0.91*0.60) = 0.036 * 0.6225 = 0.02241$.

Denominator contribution for L=F:

$0.92 * 0.45 * (0.09*0.40 + 0.91*0.15) = 0.414 * 0.1725 = 0.071415$.

$P(F=T, D=T) = 0.02241 + 0.071415 = 0.093825$.

$P(L=T \mid F=T, D=T) = 0.02241/0.093825, 0.239$; $P(L=F \mid \dots) 0.761$.

3. $P(\text{Wind}=T \mid \text{WildFire}=T, \text{SpreadRapidly}=F)$

From SpreadRapidly CPT with WildFire=T: $P(S=F|W=T, F=T)=0.10$,

$P(S=F|W=F, F=T)=0.35$.

The marginal over (L,D,H) cancels in the ratio:

$P(W=T|F=T, S=F)$ proportional to $P(W=T)*0.10 = 0.034$;

$P(W=F|F=T, S=F)$ proportional to $P(W=F)*0.35 = 0.231$.

Normalize: $P(W=T|...) = 0.034/0.265 \sim 0.128$; $P(W=F|...) : 0.872$.

4. $P(\text{OtherFire}=T \mid \text{SmokeReported}=T, \text{WildFire}=F)$

$P(O=T|M=T, F=F)$ proportional to $P(M=T|F=F, O=T)P(O=T) = 0.88 * 0.27 = 0.2376$.

$P(O=F|...)$ proportional to $0.02 * 0.73 = 0.0146$.

Sum = 0.2522. $P(O=T|...) = 0.2376/0.2522 \sim 0.942$; $P(O=F|...): 0.058$.

5. $P(\text{SpreadRapidly} \mid \text{SmokeReported}=T)$

$P(S=T|M=T) = P(S=T, M=T) / P(M=T)$.

First, $P(\text{WildFire}=T) = \text{sum over 8 (L,D,H) rows of } P(L)P(D)P(H)P(F=T|L,D,H) :$

0.113742.

WildFire and OtherFire are marginally independent, so:

$P(M=T) = P(O=T)P(F=T)*0.97 + P(O=F)P(F=T)*0.80 + P(O=T)P(F=F)*0.88 +$

$P(O=F)P(F=F)*0.02 \sim 0.319729$.

$P(S=T, M=T) = \text{sum over all (L,D,H,W,O,F) of}$

$P(L)P(D)P(H)P(W)P(O)P(F|L,D,H)P(S=T|W,F)P(M=T|F,O)$, 0.070794.

Ratio: $P(S=T|M=T) \sim 0.070794/0.319729 \sim 0.221$; $P(S=F|M=T) \sim 0.779$.

6. $P(\text{WildFire}=T)$

Sum over (L,D,H) of $P(L)P(D)P(H) P(F=T|L,D,H)$. Eight products (L,D,H each T/F):

$$(T,T,T): 0.08*0.45*0.09*0.85 = 0.002754$$

$$(T,T,F): 0.08*0.45*0.91*0.60 = 0.019656$$

$$(T,F,T): 0.08*0.55*0.09*0.50 = 0.001980$$

$$(T,F,F): 0.08*0.55*0.91*0.30 = 0.012012$$

$$(F,T,T): 0.92*0.45*0.09*0.40 = 0.014904$$

$$(F,T,F): 0.92*0.45*0.91*0.15 = 0.056511$$

$$(F,F,T): 0.92*0.55*0.09*0.12 = 0.0054648$$

$$(F,F,F): 0.92*0.55*0.91*0.001 = 0.00046046$$

Total $P(\text{WildFire}=T) \sim 0.113742$; $P(\text{WildFire}=F) \sim 0.886$